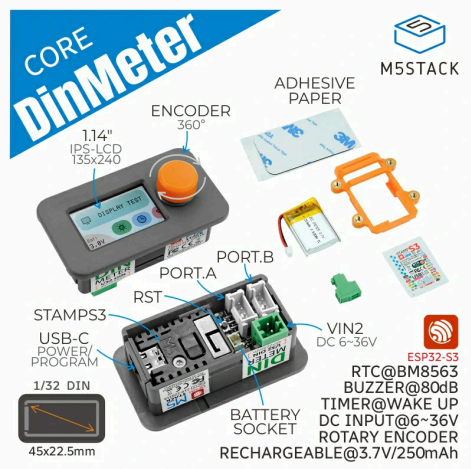
DinMeter

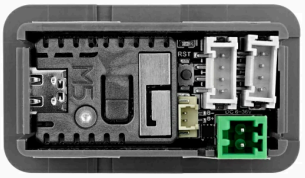
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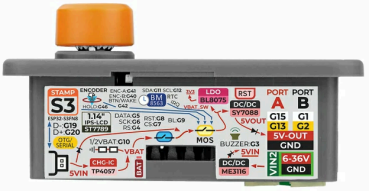














Description

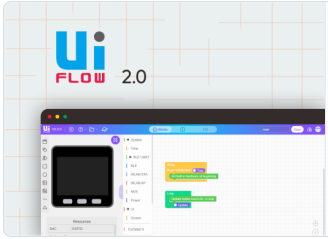
DinMeter is an embedded development board that adheres to the 1/32DIN standard, featuring a 1.14-inch ST7789 screen and a built-in rotary encoder with buttons for precise knob position recording. Additionally, it includes an RTC circuit, an onboard buzzer, and buttons for device interaction and wake-up alerts. The product is designed to support DC input ranging from 6V to 36V and includes a lithium battery interface and charging circuit to accommodate various power supply needs. The reserved PORT.A and PORT.B interfaces allow for easy expansion of I2C and GPIO devices. This product is suitable for applications such as parameter measurement and detection, smart home control, IoT projects, smart wearables, access control, industrial control, and educational maker projects.

Tutorial



Arduino IDE

This tutorial will introduce how to program and control the DinMeter device using the Arduino IDE.



UiFlow2

This tutorial will introduce how to control the DinMeter device through the UiFlow2 graphical programming platform.

Features

- ST7789v2 screen
- Stamp-S3 as the main controller
- Rotary encoder
- Wide voltage input
- Reserved PORTA and PORTB interfaces
- Lithium battery interface (rechargeable)
- 1/32DIN standard (45 x 22.5mm)
- Development Platform
  - UiFlow2
  - Arduino IDE
  - ESP-IDF
  - PlatformIO

Includes

- 1 x DinMeter
- 1 x Adhesive tape
- 1 x 250mAh polymer lithium battery
- 1 x Silk-screened sticker
- 1 x DC-JACK terminal input (6-36V)
- 1 x Back bracket

Applications

- Smart home detection and control systems
- Access control systems
- Industrial control
- Maker and DIY projects

Specifications

Specification	Parameter
SoC	ESP32-S3FN8 @ Xtensa® 32-bit LX7 dual-core processor, clock frequency 240MHz
Flash	8MB
Wi-Fi	2.4 GHz Wi-Fi
USB	USB OTG, USB Serial/JTAG
Wide Voltage Input Range	6 ~ 36V
Display Driver	ST7789V2
Display Resolution	135 x 240
Charging Current	100mA
Grove Load Capacity	PORT.A Load Capacity: DC 5V/Max:220mA PORT.B Load Capacity: DC 5V/Max:220mA
Standby Current (Battery Powered)	DC 4.2V@38.4uA
Battery Socket Specification	MX1.25-2P
Power Supply Method	USB/DC Power Supply / Battery
Operating Temperature	0 ~ 40°C
Product Size	53.0 x 30.0 x 32.0mm
Product Weight	17.8g
Package Size	136.0 x 93.0 x 23.0mm
Gross Weight	47.0g

Learn

Power On/Off Operation

Power On: The device can be woken up by pressing the "WAKE" button or by an RTC-triggered IRQ signal. After the wake-up signal is triggered, the program initialization must set the hold(G46) pin to high (1) to maintain power; otherwise, the device will re-enter sleep mode.

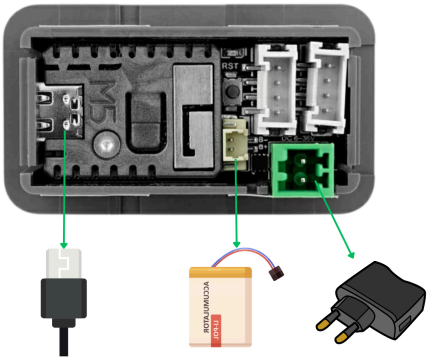
Power Off: When there is no external USB power supply, press the RST button to power off, or set HOLD(GPIO46)=0 in the program to power off.

Download Mode

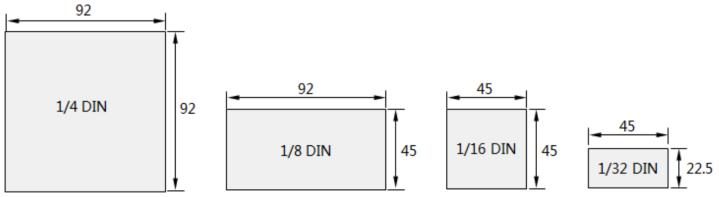
To enter download mode, hold the G0 button on the StampS3 before powering on, then release it after powering up.



Power Supply Methods | Three power supply methods: USB/DC/BAT



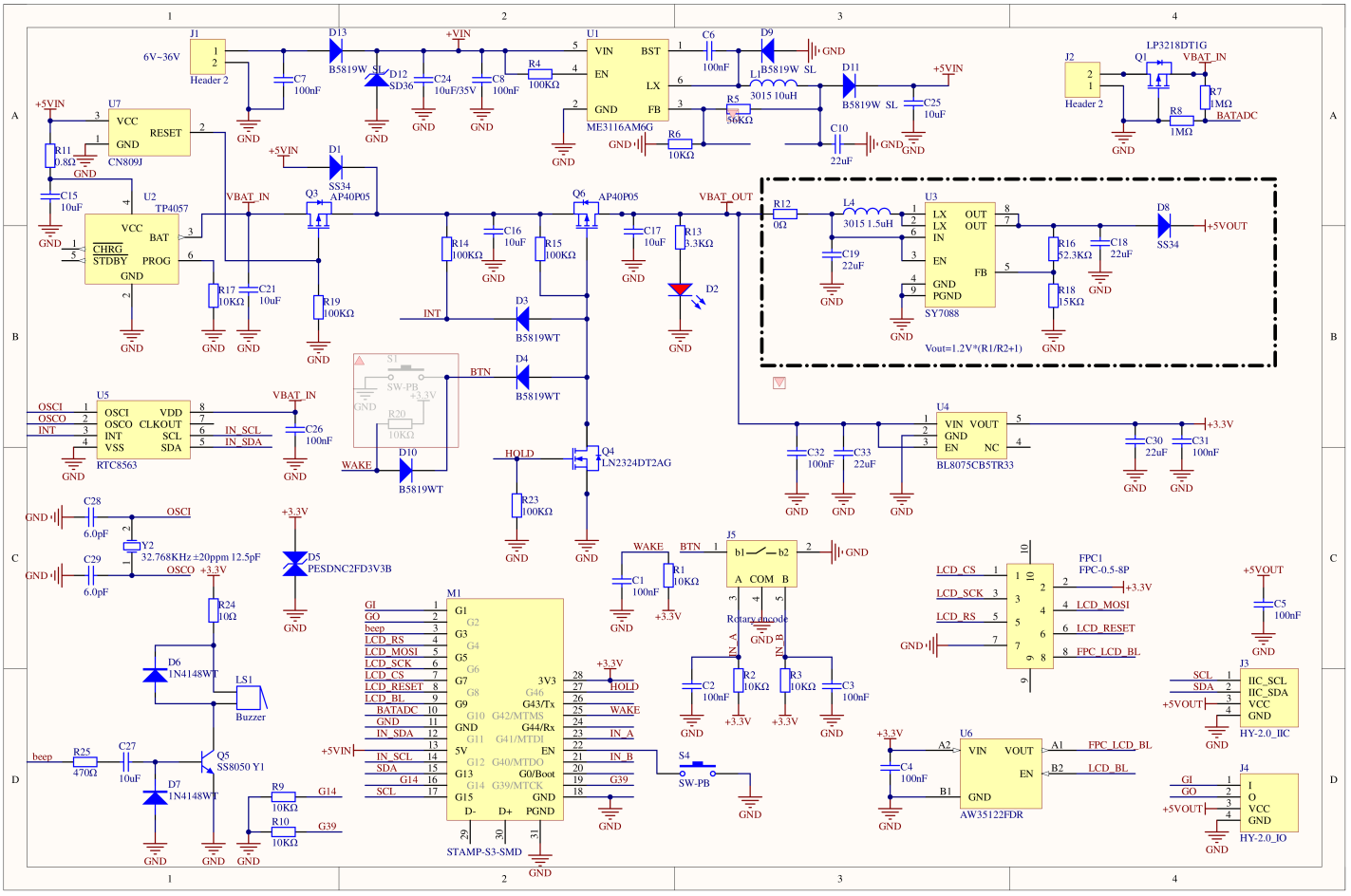
Din Size Standard



Panel Cutouts : Standard Industrial Size

Schematics

- DinMeter Schematics PDF



PinMap

RTC8563 & ENCODER

ESP32-S3FN8	G12	G11	G40	G41	G3	G42
RTC8563	SCL	SDA				
ENCODER			B	A		
BEEP					beep	
BUTTON (ENCODER)						BTN(WAKE)

ST7789V2

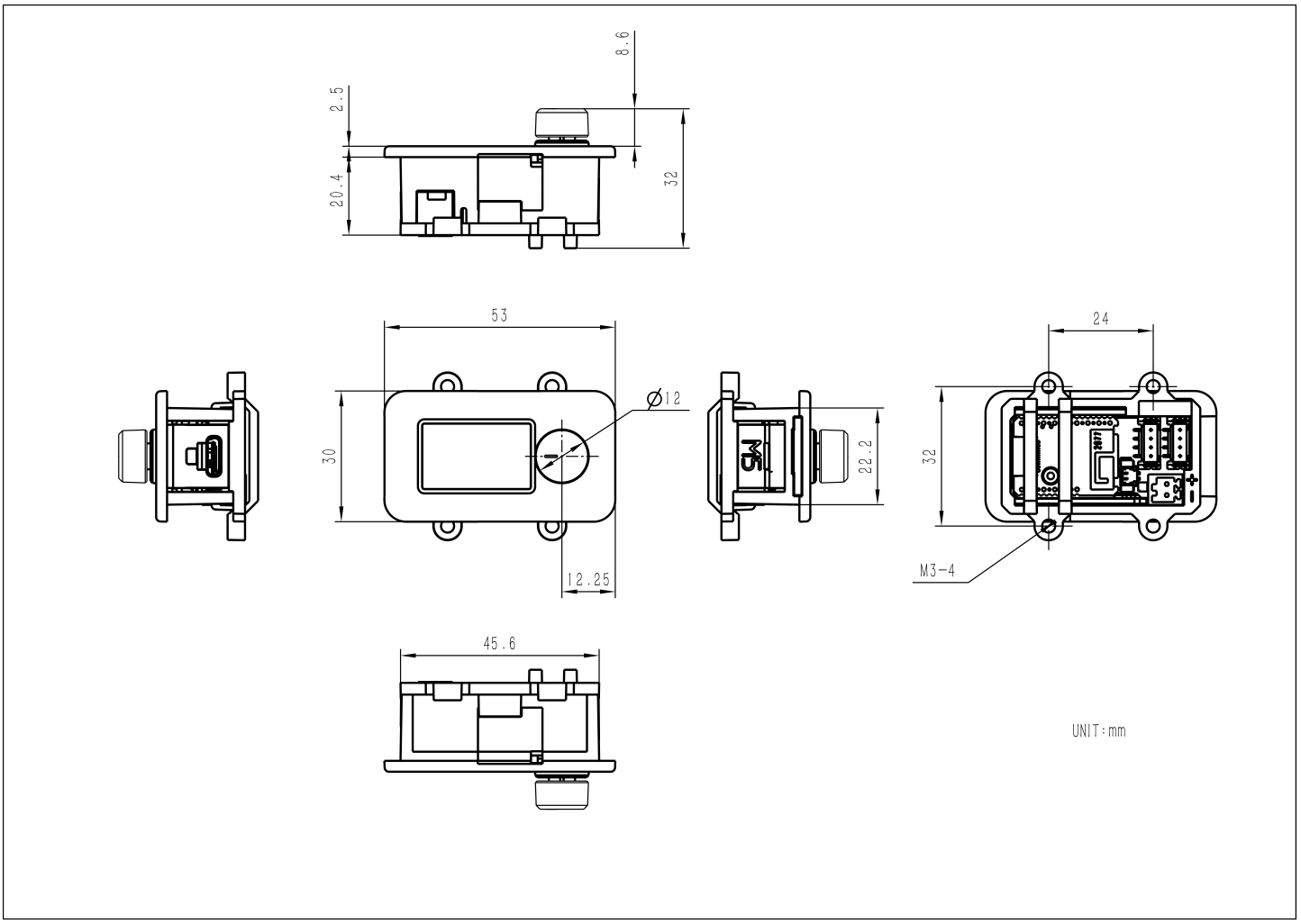
ESP32-S3FN8	G7	G6	G4	G5	G8	G9
ST7789V2	CS	SCK	RS	MOSI	RESET	BL

HY2.0-4P

HY2.0-4P	Black	Red	Yellow	White
PORT.A	GND	5V	G13	G15
PORT.B	GND	5V	G2	G1

Model Size

- DinMeter Model Size PDF



Structure

- DinMeter Structure Files

Datasheets

- BM8563
- tp4057
- ST7789V2
- Spec of external DC power connector
- Lithium battery seat

Softwares

Quick Start

- DinMeter Arduino Quick Start

Arduino

- DinMeter Arduino Driver Library
- DinMeter Factory Firmware

UiFlow2

- DinMeter UiFlow2 Quick Start

Easyloader

Easyloader	Download Link	Notes
DinMeter User Demo Easyloader	download	/

Video